A Difference-in-Difference Approach to Detecting AI-Generated Images

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Abstract

Diffusion models are able to produce AI-generated images that are almost indistinguishable from real ones. This raises concerns about their potential misuse and poses substantial challenges for detecting them. Many existing detectors rely on reconstruction error — the difference between the input image and its reconstructed version — as the basis for distinguishing real from fake images. However, these detectors become less effective as modern AI-generated images become increasingly similar to real ones. To address this challenge, we propose a novel difference-in-difference method. Instead of directly using the reconstruction error (a first-order difference), we compute the difference in reconstruction error – a second-order difference – for variance reduction and improving detection accuracy. Extensive experiments demonstrate that our method achieves strong generalization performance, enabling reliable detection of AI-generated images in the era of generative AI. Code is available at https://github.com/Qixinyi1122-lucky/DID.

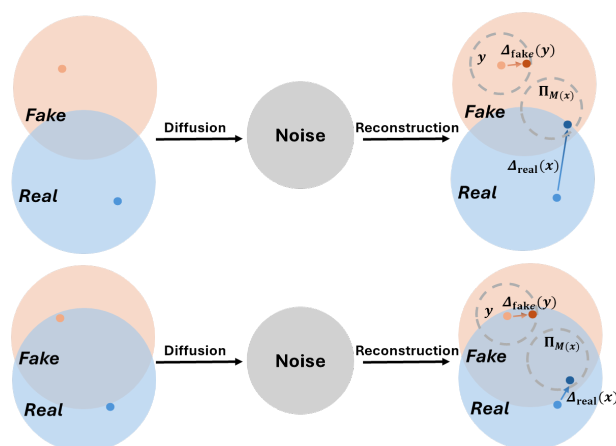

Figure 1. **Left:** The real image x and its distribution on the manifold $\mathcal{M}$ are shown in blue, whereas the synthetic image y and its distribution are shown in orange. **Right:** Reconstructions of the real and synthetic images. Δ_{fake} and Δ_{real} denote the reconstruction errors of synthetic and real images, respectively, and $\Pi_{\mathcal{M}}$ represents the projection of the real image onto $\mathcal{M}$. **Top:** The generator is weak, resulting in a large discrepancy between the real and synthetic image distributions. Consequently, the reconstruction error of the synthetic image, $\Delta_{\text{fake}}(y)$, is much larger than that of the real image, $\Delta_{\text{real}}(x)$, making detection easier. **Bottom:** The generator is strong, so the real and synthetic image distributions are close. Here, $\Delta_{\text{fake}}(y)$ is similar to $\Delta_{\text{real}}(x)$, making AI-generated images difficult to detect.

1. Introduction

The rapid evolution of large-scale generative models has led to groundbreaking progress in image generation in recent years [2, 4, 6, 9, 69] . Diffusion models, in particular, have become the foundation of modern image generation. However, the proliferation of AI-generated images has raised substantial concerns about authenticity and public trust, creating an urgent need for effective detectors that can distinguish AI-generated images from real ones. There is a growing literature on detecting AI-generated images; see Section 2 for a detailed review. Early detection algorithms were tailored for specific models and exploited model-based artifacts — such as inconsistencies in the frequency domain,

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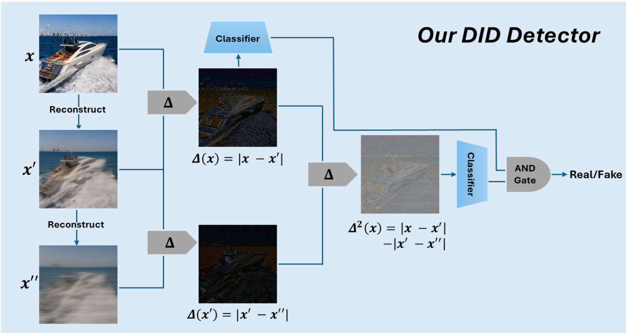

Figure 2. Visualizations of our DID detector. Δ denotes the differencing operator that computes the pixel-wise difference between two images.

irregularities in color filter responses, or statistical traces introduced by up-sampling layers [20, 27, 94]. Although these methods achieved strong performance in detecting images from specific generators, they lacked robustness across different models.

More recently, reconstruction-based algorithms have emerged as a promising class of detectors that are both conceptually intuitive and practically effective. As suggested by their name, these methods (a) reconstruct the image to be detected using a generative model (e.g., a diffusion model), (b) compute the reconstruction error as the difference between the original and reconstructed images, and (c) classify the image as real or synthetic based on that error. Their core idea is straightforward: given a particular generative model for reconstruction, synthetic images that lie on or near the model manifold should be reconstructed with smaller residual errors than real images, which reside outside that manifold (see the upper panel of Figure 1 for an illustration). This line of work, represented by DIRE [80] and its variants LaRE^2 and FIRE [18, 49], has generalized well across different generative models, while maintaining a transparent interpretation.

However, the effectiveness of existing reconstruction-based algorithms relies on a sufficiently large gap between real and AI-generated images to be detectable. As generative models advance rapidly, this assumption becomes increasingly invalid. Meanwhile, images found on the internet are often post-processed, through resizing, compression, or partial AI editing (where only specific regions are synthesized), which further confounds detection signals. Recent studies have empirically observed that as the quality of the image generator improves or under adversarial scenarios (e.g., with partial AI edits), reconstruction-based algorithms struggle to detect AI-generated content effectively [18, 79]. See also, the bottom panel of Figure 1 for an illustration. Therefore, as high-fidelity diffusion models become prevalent, it is necessary to move beyond existing reconstruction-based methods and develop more effective algorithms capable of detecting subtler differences between real and AI-generated content.

This paper proposes a new algorithm, termed *difference-in-differences* (DID), for detecting AI-generated content. Instead of relying solely on the reconstruction error – which can be viewed as a *first-order* difference between the input image and its reconstruction – we compute a *second-order* difference by taking the difference between the reconstruction error of the input image and the reconstruction error of its reconstructed version. Our analytical analysis in Section 3 shows that this second-order differencing can effectively remove fluctuations introduced during synthetic image generation, thus amplifying the detection signal particularly when it is weak. We next employ both first-order and second-order differences as the basis for detection (see Figure 2 for an illustration of our proposal). Empirically, our extensive numerical experiments demonstrate that DID consistently outperforms existing detectors across diverse combinations of image datasets and generative models, achieving improvements around 20% - 30% over the strongest baseline. Moreover, our findings shed light on the effectiveness of DID: in simpler settings where AI-generated images are sufficiently different from real ones, first-order differences alone are adequate for detection; however, in more complex experimental settings, employing the second-order difference substantially improves the detection accuracy. These observations are consistent with our analytical insights. By leveraging both orders of differences, our method generalizes robustly across a wide range of detection tasks.

2. Related Works

This section is organized as follows. We begin by reviewing various diffusion models used for image generation. We then review existing algorithms for AI-image detection. Finally, we introduce the difference-in-difference (DID) method from econometrics, which is conceptually related to our proposal but fundamentally different in terms of its problem setting and application.

2.1. Diffusion Models

Diffusion models have become a cornerstone of modern image generation and multimodal large language models [MLLMs, 35, 89], powering a wide range of tasks such as text-to-image, text-to-3D generation, and text-to-video generation [22, 30, 33, 43, 55, 59, 62, 63, 66, 67, 88, 91]. They constitute an important family of probabilistic generative models that progressively perturb data with noise and learn a reverse process to generate synthetic samples [86]. Broadly speaking, these models can be categorized into three main types: denoising diffusion probabilistic models [32, 70], score-based generative models [72, 73], and stochastic differential equation (SDE)-based models

[74, 75]. Building on the three types of models described above, each can be further decomposed into two subtypes: unconditional and conditional diffusion models. For unconditional diffusion models, prior research has focused on three directions: (i) accelerating the sampling process [25, 38, 72, 81, 92, 93], (ii) tightening the variational lower bound on the data log-likelihood [7, 39, 48, 56, 71], and (iii) extending the model to accommodate new data modalities [12, 34, 37, 57, 77]. By contrast, research on conditional diffusion models has focused on guiding the generation process to produce content that aligns with users' desired specifications. For instance, ADM uses an auxiliary classifier to improve the quality of generated samples [24]. ImageReward [84] and Diffusion-DPO [78] leverage reinforcement learning from human feedback to align diffusion models with human preferences. Scalable diffusion models [60] replace standard layer normalization in transformers with adaptive layer normalization to more effectively encode conditioning variables. DiffuSeq [28], unCLIP [63], ConPreDif [85], and RPG [87] focus on guiding diffusion models with textual inputs.

2.2. AI-Generated Image Detection

With the rapid advancement of AI image generators, there is an urgent need to detect AI-generated images; see [45] for a recent review of detection algorithms.

Early works leverage physical [11, 26, 51, 52] and physiological features [19] in the images, such as inconsistencies in color, perspective, shadows, reflections or human anatomy for detection. These methods rely on the assumption that AI-generated images often fail to adhere to real-world physical and physiological constraints. Later approaches employ spatial-domain features, such as the image texture to build classifiers [47, 54, 95]. However, such classifiers often fail to capture frequency-domain artifacts [82]. To overcome this limitation, several methods have been proposed that combine spatial- and frequency-domain features for detection [5, 40, 82, 83]. Other approaches leverage alternative features, such as the negative log-likelihood of the input image [21], or directly fine-tune an MLLM [44] for detection.

Different from the aforementioned approaches, our proposal is most closely related to reconstruction-based detection algorithms. As discussed in the introduction, a representative example is DIRE [80], which employs a diffusion model with a forward diffusion process and a reverse denoising process to obtain the reconstruction, computes the difference from the original image as the reconstruction error, and trains the classifier using this reconstruction error. SeDID [50] extends this approach by considering not only the final reconstruction error between the reconstructed and input image, but also the intermediate errors accumulated along the forward diffusion and reverse denoising processes. Other variants, such as DRCT [17], modify the objective function by employing a contrastive loss to train the classifier based on the reconstruction error. One potential limitation of DIRE is its computational time required to obtain the reconstruction, as the forward diffusion and reverse denoising processes involve multiple steps and can be time-consuming. To address this, LaRE^2 [49] projects the image into a latent space and applies the forward and reverse procedures only once, computing the reconstruction error in the latent space to speed up the computation. AEROBLADE [64] adopts a training-free strategy: instead of training a classifier with the reconstruction error, it relies on simple statistical measures derived from the reconstruction error for detection. FakeInversion [16] inverts an open-source stable diffusion process model to compute the reconstruction error. Finally, FIRE additionally leverages frequency-domain information to compute reconstruction errors, which enhances the detector's robustness in adversarial settings and across different generative models [18].

2.3. Difference-in-Differences

Our proposal is conceptually similar to the difference-in-difference approach for causal inference, which estimates the treatment effect of a particular policy. DID originated in social science research on quasi-experimental designs [14, 36] and was later systematized and popularized in the econometrics literature for analyzing panel data with multiple subjects observed over time [3, 10, 15].

In the typical DID setting, some subjects are continuously exposed to a control policy, while others switch from the control to a new policy at a certain time point. For the latter group, one can compute a first-order difference in their average outcomes before and after the policy change. However, this difference captures not only the effect of the policy but also any time-related effects. To remove the confounding effect of time, a corresponding first-order difference is computed for the control group, and a second-order difference between the two first-order differences is then taken to isolate the treatment effect. For more details, refer to [1, 13, 29, 76].

Despite sharing the same name and similar conceptual ideas, our proposal is designed for a fundamentally different application of AI-generated image detection, as opposed to causal inference.

3. Methodology

This section first provides background on diffusion models (Section 3.1), then discusses the intuition and limitations of existing reconstruction-based methods (Section 3.2), which motivates our use of DID (Section 3.3). Finally, we detail our proposal (Section 3.4).

3.1. Background on Diffusion Models

Image generation via diffusion models typically consists of two processes: a *forward diffusion* process that gradually corrupts data with Gaussian noise, and a *reverse denoising* process that reconstructs the underlying image signal from noise. Specifically, in the forward process, an image x_0 is progressively transformed into a sequence of noisy images $\{x_t\}_{t=1}^T$ according to

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\bar{\alpha}_t}\,\epsilon_t,$$

for $t = 1, \ldots, T$, where x_t denotes the noisy image at timestep t, and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ for a sequence of pre-defined weights $\{\alpha_t\}_t$ that controls the signal-to-noise ratio and $\epsilon_t \sim \mathcal{N}(0, I)$ is the Gaussian noise.

In contrast, during the reverse process, the model learns to iteratively denoise x_t back to a clean sample using a parameterized neural network $\epsilon_\theta(x_t, t)$ that takes x_t and t as input and aims to predict the added noise ϵ_t. This process can be mathematically represented by

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(x_t, t)\right) + \sqrt{1-\alpha_{t-1}}\,\epsilon_t,$$

for $t = T, \ldots, 1$, where $\epsilon_\theta(x_t, t)$ is trained to minimize the prediction error between the true and the predicted noise.

3.2. Intuition and Limitations in Reconstruction-Based Detection

Let $\mathcal{X} \subseteq \mathbb{R}^{3\times 256\times 256}$ denote the space of all natural images, and $\mathcal{M}$ represent the manifold of images generated by a given pre-trained diffusion model. In general, $\mathcal{M}$ does not coincide with $\mathcal{X}$, as the model's learned distribution cannot perfectly align with the true image distribution. However, as generative models become increasingly expressive, $\mathcal{M}$ tends to approximate $\mathcal{X}$ more closely, forming a high-fidelity submanifold near the real image space.

The reconstruction can thus be interpreted as a projection of an input image x onto the generative manifold $\mathcal{M}$, followed by a stochastic perturbation[1]:

$$\mathcal{R}(x) = \Pi_{\mathcal{M}}(x) + \delta(x), \tag{1}$$

where $\mathcal{R}(\bullet)$ denotes the reconstruction operator, $\Pi_{\mathcal{M}}(x)$ denotes the projection of x onto $\mathcal{M}$, and $\delta(x)$ is a perturbation term capturing the randomness introduced during the reconstruction.

Consequently, for a synthetic image $x \in \mathcal{M}$, its projection onto $\mathcal{M}$ equals x itself, resulting in a reconstruction error of

$$\Delta_{\text{fake}}(x) = |\delta(x)|, \tag{2}$$

where the absolute value is taken elementwise. In contrast, for a real image $x \in \mathcal{X}$, its reconstruction error is given by

$$\Delta_{\text{real}}(x) = |x - \Pi_{\mathcal{M}}(x) - \delta(x)|. \tag{3}$$

When the generative model is weak, the discrepancy between $\mathcal{M}$ and $\mathcal{X}$ is large, so $|x - \Pi_{\mathcal{M}}(x)|$ dominates the reconstruction error. As a result, the real image achieves a much larger reconstruction error than the synthetic image (see the upper panel of Figure 1 for an illustration), and we classify an image as synthetic if its reconstruction error is small. This explains the rationale behind existing reconstruction-based methods. When the generative model is strong, however, the real and synthetic image distributions are closely aligned, so the signal $|x - \Pi_{\mathcal{M}}(x)|$ becomes small. Consequently, the reconstruction errors of both real and synthetic images are dominated by the perturbation and become similar (see the bottom panel of Figure 1), and existing reconstruction-based methods struggle to distinguish real from synthetic images.

In summary, existing reconstruction-based methods are limited in settings where synthetic images closely resemble real ones, due to the strong capacity of modern generative models. This indicates that relying solely on first-order differences is insufficient in such challenging scenarios, which motivates our difference-in-difference approach that seeks higher-order differences to remove perturbation errors while amplifying the weak signal. We will demonstrate how DID addresses this challenge in the next section.

3.3. Difference-In-Differences

Unlike existing methods which conduct reconstruction only once, DID performs two consecutive reconstructions using the same pre-trained diffusion model:

$$x' = \mathcal{R}(x), \quad x'' = \mathcal{R}(x'),$$

where $\mathcal{R}(\bullet)$ is the reconstruction operator defined in (1), x' denotes the reconstruction of the input image x and x'' denotes the reconstruction of the reconstructed image.

Taking the difference between x' and x yields the first-order reconstruction error, $\Delta(x) = |x - \mathcal{R}(x)|$, which forms the basis of existing reconstruction-based detectors for classification. However, as discussed in the previous section, relying solely on the first-order reconstruction error is limited when the signal $|x - \Pi_{\mathcal{M}}(x)|$ for a real image is small, causing the reconstruction error to be dominated by the perturbation $\delta(x)$. To address this limitation, DID computes the second-order difference between the two reconstruction errors:

$$\Delta^2(x) = |x - x'| - |x' - x''|. \tag{4}$$

To illustrate the intuition behind DID, note that, similar to (2) and (3), one can derive that for a synthetic image x, the

[1]Similar interpretations have been proposed in the context of LLM-generated text detection [97].

second-order reconstruction error is given by

$$\Delta^2_{\text{fake}}(x) = |\delta(x)| - |\delta(x')|, \tag{5}$$

whereas for a real image, it is

$$\Delta^2_{\text{real}}(x) = |x - \Pi_{\mathcal{M}}(x) - \delta(x)| - |\delta(x')|. \tag{6}$$

When the signal is weak, x is close to x'. In this case, if the perturbation error δ is highly correlated across space, then $\delta(x) \approx \delta(x')$. As a result, the two perturbations cancel out, and (5) and (6) simplify to

$$\Delta^2_{\text{fake}}(x) \approx 0, \tag{7}$$

and

$$|\Delta^2_{\text{real}}(x)| = |x - \Pi_{\mathcal{M}}(x) - \delta(x) + \delta(x')| \approx |x - \Pi_{\mathcal{M}}(x)|, \tag{8}$$

where the first equality holds because the weak signal makes the magnitude of $|x - \Pi_{\mathcal{M}}(x)|$ much smaller than that of $|\delta(x)|$, so that the sign of $x - \Pi_{\mathcal{M}}(x) - \delta(x)$ is dominated by that of $-\delta(x)$.

Comparing (7) and (8) with (2) and (3), it becomes clear that, unlike the first-order difference, the second-order difference is no longer confounded by the perturbation error and depends only on the signal, even when it is weak. This highlights the advantage of DID over existing reconstruction-based algorithms: by taking a second-order difference, it effectively removes the perturbation noise and uncovers the subtle signal.

3.4. Our Proposal

We present the details of our proposed method in this section. Based on the aforementioned analysis, the first-order reconstruction error $\Delta(x) = |x - x'|$ is effective for distinguishing real from synthetic images when their discrepancy is large, whereas the second-order reconstruction error $\Delta^2(x) = |x - x'| - |x' - x''|$ remains reliable when real and synthetic images are highly similar. To handle both cases simultaneously, our detector integrates both first- and second-order reconstruction errors for robust detection; see Figure 3 and Section D for visualizations of these errors.

Specifically, we train two classifiers independently based on $\Delta(x)$ and $\Delta^2(x)$, respectively, using the standard cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(y_i') + (1 - y_i)\log(1 - y_i')\right], \tag{9}$$

where y_i denotes the ground-truth label indicating whether the image is real or synthetic, y_i' is the predicted probability produced by the classifier, and N is the number of training samples.

Each classifier outputs a numerical probability score and predicts an image as real if the score is smaller than a predefined threshold c. Finally, an image is classified as real only when both classifiers identify it as real.

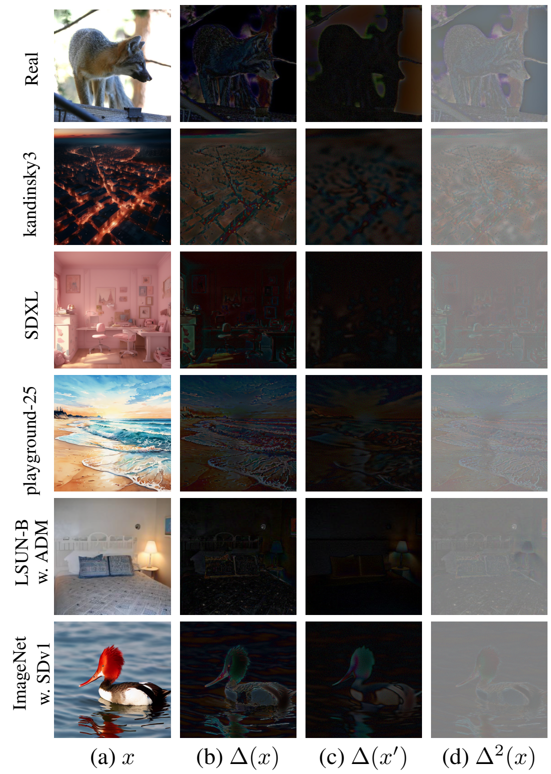

Figure 3. Visualizations of (a) input image x; (b) its first-order reconstruction error $\Delta(x)$; (c) the reconstruction error of the reconstructed image x'; and (d) the second-order reconstruction error $\Delta^2(x)$. Compared to the first-order error $\Delta(x)$, the second-order error $\Delta^2(x)$ better distinguishes real from fake images: real images produce more stable and brighter responses, while fake images exhibit noticeably weaker signals, resulting in a larger gap between the two categories.

4. Experiments

In this section, we conduct extensive numerical experiments to investigate the finite-sample performance of the proposed detector. We begin by describing the experimental setting in Section 4.1. We next compare our detector against existing state-of-the-art detectors in Section 4.2. Finally, we conduct a sensitivity analysis and an ablation study to examine the robustness of our detector and the contribution of the second-order difference in Sections 4.3 and 4.4, respectively.

4.1. Experimental Setup

To comprehensively evaluate the effectiveness of each detector, we consider a variety of datasets and generative models. Our evaluation benchmark closely follows those used in DIRE [80], FIRE [18], and FakeInversion [16]. Specifically, we use two datasets – ImageNet [23] and LAION [68] – to train the classifier for each detector, with one large-scale training set (40k real and 40k synthetic images) and one

Table 1. Accuracy (%) of different detectors when trained on **ImageNet & ADM**. The best results are highlighted in **bold**, while the second best results are <u>underlined</u>. All averages are computed from the raw values and then rounded to two decimals.

Train set & Gen. model		**ImageNet & ADM**				
Eval set	**Gen. model**	**DID**	**DIRE**	**LaRE^2**	**AERO.**	**UFD**
LSUN-B	ADM	**99.5**	**99.5**	50.0	52.3	<u>63.9</u>
	PNDM	**97.6**	**97.6**	55.4	49.5	<u>91.8</u>
	DDPM	<u>99.5</u>	**99.8**	56.6	56.7	<u>76.5</u>
	iDDPM	<u>99.4</u>	**99.6**	50.0	50.3	85.1
	SDv1	<u>99.7</u>	<u>99.9</u>	**100**	50.6	94.5
	Avg.	<u>99.12</u>	**99.25**	62.39	51.87	82.36
ImageNet	ADM	**100**	**100**	**100**	51.0	<u>85.8</u>
	SDv1	**99.3**	98.9	<u>99.2</u>	37.0	79.9
	Avg.	**99.64**	99.44	<u>99.60</u>	43.99	82.80
LAION	Kan.3	**99.5**	<u>99.3</u>	64.5	66.7	72.6
	SDXL	**99.6**	<u>99.4</u>	66.4	55.5	69.5
	Vega	**99.7**	<u>99.5</u>	66.4	52.2	68.6
	Playground v2.5	**99.8**	**99.8**	66.0	53.1	<u>82.8</u>
	Stable Cascade	**99.6**	<u>99.4</u>	66.3	73.4	81.4
	Avg.	**99.60**	<u>99.45</u>	65.89	60.14	75.95
Total average		**99.41**	<u>99.37</u>	70.05	54.00	79.35

smaller set (10k real and 10k synthetic images). For ImageNet, we randomly sample 40,000 real images and 40,000 synthetic images generated by ADM [24]. For LAION, we randomly sample 10,000 real images and generate 10,000 synthetic images using open-source text-to-image models such as Kandinsky 3.0 [2] and SDXL [62], with prompts collected from Midjourney.

For testing, it is crucial to evaluate each detector on unseen datasets in order to assess its generalization ability. In the main paper, we report results on three test datasets—LAION [68], ImageNet [23], and LSUN-Bedroom [90], the last of which is excluded from training. For these evaluations, synthetic images are generated using a more diverse set of generative models than those used during training, including ADM [24], PNDM [46], DDPM [32], iDDPM [56], SDv1 [65], Segmind Vega [31], Playground v2.5 [41], and Stable Cascade [61]. In addition, Section C.2 includes further evaluations on additional datasets, such as GenImage [100] and GenDet [99], as well as a broader set of generative models. More details on dataset construction and experimental setup are provided in Section A.

We use the ADM model pretrained on LSUN-Bedroom to obtain the reconstructed image. After obtaining the two reconstructions, both the first- and second-order differences are fed into a ResNet-50 classifier for detection. For all baseline detectors, the classification threshold is fixed at the default level $c = 0.5$. Our method, however, uses two classifiers and labels an input as real only when both classifiers predict it as real. To allow for a fair comparison, we set the threshold of each classifier in DID to $1 - \sqrt{0.5} \approx 0.29$, so that its overall false positive rate matches that of using a single classifier with a threshold of 0.5. Accuracy (ACC) is reported as the primary evaluation metric. All our experiments are run on NVIDIA H800 GPUs ; the computation and memory cost is reported in Section C.3.

4.2. Comparision against Existing Baselines

In the main paper, we compare our proposal against the following four state-of-the-art algorithms. These baseline methods include four reconstruction-based algorithms – (i) **DIRE** [80], (ii) **LaRE**^{2} [49], and (iii) AEROBLADE [64] (denoted as **AERO.**) – as well as (iv) a non–reconstruction-based approach, UniversalFakeDetect (denoted as **UFD**) [58], which leverages a feature space not explicitly trained to distinguish real from synthetic images. Further details are provided in Section B; comparisons aganist additional baseline algorithms are reported in Section C.2.

Performance with a Larger Training Dataset. When trained on the ImageNet dataset with 40k real images and 40k synthetic images generated by ADM, we report the accuracy of different detectors in Table 1. DID achieves performance very similar to DIRE. This is expected for two reasons. First, the training dataset contains a large number of samples. Second, we also use ADM for reconstruction, so the generative model used to produce the synthetic training images aligns with the one used for reconstruction. Under this large-sample and well-aligned setting, the first-order reconstruction error is already highly discriminative for distinguishing real from AI-generated images. As a result, both DID and DIRE achieve classification accuracy close to 100%. In contrast, other methods, such as AER-

Table 2. Accuracy (%) of different detectors when trained on **LAION & Kan.3** (left block) and **LAION & SDXL** (right block). The best results are highlighted in **bold**, while the second best results are underlined. All averages are computed from the raw values and then rounded to two decimals. The last column of each block reports the improvement of DID over the second-best baseline.

Train set & Gen. model		LAION & Kan.3					Imp.	LAION & SDXL					Imp.
Eval set	**Gen. model**	**DID**	**DIRE**	**LaRE^2**	**AERO.**	**UFD**		**DID**	**DIRE**	**LaRE^2**	**AERO.**	**UFD**	
LSUN-B	ADM	**90.7**	88.1	50.3	50.0	50.8	21.85%	**95.0**	94.3	50.2	50.0	50.3	13.04%
	PNDM	**78.2**	73.5	71.2	49.8	51.5	17.55%	**81.7**	74.6	51.6	49.8	50.9	27.95%
	DDPM	**88.7**	85.2	56.5	56.4	51.5	24.05%	**95.1**	93.3	57.1	56.4	50.5	27.73%
	iDDPM	**80.1**	77.5	50.0	49.8	54.2	11.33%	**89.0**	86.8	50.0	49.8	53.3	16.29%
	SDv1	98.6	97.8	**99.9**	49.8	49.7	-	98.4	98.3	**99.7**	49.8	50.5	-
	Avg.	**87.25**	84.41	65.56	51.13	51.52	18.23%	**91.83**	89.44	61.70	51.13	51.09	22.58%
ImageNet	ADM	**99.1**	96.7	89.5	50.7	58.7	74.03%	**99.6**	96.3	93.3	50.7	61.1	89.65%
	SDv1	**99.2**	97.7	94.4	35.0	65.4	65.33%	**99.1**	96.1	95.5	35.0	67.3	75.73%
	Avg.	**99.16**	97.16	91.91	42.83	62.03	70.46%	**99.34**	96.22	94.40	42.83	64.18	82.47%
LAION	Kan.3	**100**	**100**	**100**	58.7	93.7	-	**99.8**	99.3	97.6	58.7	92.9	64.29%
	SDXL	**100**	99.9	99.4	52.5	87.0	100%	**99.9**	**99.9**	**99.9**	52.5	92.3	-
	Vega	**100**	99.4	95.5	51.0	79.9	100%	**99.9**	99.8	99.7	51.0	83.8	50.00%
	Playground v2.5	**100**	**100**	97.9	51.1	93.2	-	**99.9**	**99.9**	99.4	51.1	93.3	-
	Stable Cascade	**100**	**100**	99.6	64.7	95.7	-	**99.8**	99.5	99.5	64.7	95.4	54.55%
	Avg.	**100**	99.84	98.45	55.57	89.87	100%	**99.84**	99.67	99.20	55.57	91.50	51.52%
Total average		**94.55**	92.96	83.66	51.60	69.25	22.52%	**96.42**	94.83	82.78	51.60	70.11	30.66%

OBLADE, LaRE^2 and UFD, may perform competitively on certain datasets but are less robust across datasets, and their accuracy generally falls behind both DIRE and our DID.

Performance with a Smaller Training Dataset. We report the accuracy of various detectors trained on the LAION dataset in Table 2. This represents a more challenging setting for two reasons: (i) the sample size is smaller, with only 10k real and 10k synthetic images; (ii) the generators used to create the synthetic training datasets (Kan3 and SDXL) differ from the ADM model used for reconstruction. In such cases, the first-order reconstruction discrepancy alone is no longer sufficiently reliable. Consequently, DID benefits from its second-order reconstruction error and outperforms DIRE in almost all settings, particularly when evaluated on LSUN-B and ImageNet, where the test images differ from the training dataset LAION. Across these cases, our total improvement ranges from 20–30%. This demonstrates that our second-order reconstruction error plays a crucial role in more complex tasks, especially when the sample size is smaller or when the generator differs from the reconstruction model. Other baseline detectors, such as LaRE^2 and UFD, perform well when the training and testing datasets are the same, but their performance drops when the datasets differ. This demonstrates their limited generalization ability. AERO. also fails in most cases.

4.3. Sensitivity Analysis

In Section 4.2, all synthetic images were generated using diffusion models. In this section, we further evaluate each detector's generalization ability by testing it on images produced by generative adversarial networks (GANs). We consider three GAN models, namely StyleGAN, ProjectedGAN, and Diff-ProjectedGAN, and report the ACC of each detector in Table 3. Although our model is trained exclusively on diffusion-generated data from "Kan.3" or "SDXL," it consistently achieves the highest accuracy across all three GAN variants. In every setting, our method outperforms DIRE, with improvements of up to 20.90%. The performance gap becomes even more pronounced when compared with LaRE^2, and AEROBLADE, all of which fail to generalize to these GAN-generated distributions. UniversalFakeDetect also struggles in this scenario. These results demonstrate that our detector – despite being trained solely on diffusion-generated images – remains robust to synthetic images produced by substantially different generative mechanisms. We also conduct another sensitivity analysis across image formats in Section C.1.

4.4. Ablation Study

In this section, we conduct an ablation study to compare our DID with a variant (denoted as Δ^2) that uses only the second-order reconstruction error and excludes the first-order term. For completeness, we also include DIRE as a reference. To facilitate the computation, we train a lightweight model using only 10% of the LAION training data and report the results in Table 4.

We make several observations. First, in most settings, Δ^2 outperforms DIRE, demonstrating that second-order differences are more effective than first-order differences at capturing subtle discrepancies between real and synthetic

Table 3. Detection accuracy (%) on GAN-generated images under different training settings. Best in **bold**, second best <u>underlined</u>. All averages are computed from the raw values and then rounded to two decimals. The last column reports the improvement of DID over the second-best baseline.

Train set & Gen. model	Eval set	Gen. model	Accuracy (%)					Imp.
			DID	DIRE	LaRE^2	AERO.	UFD	
LAION & Kan.3	LSUN-B	StyleGAN	**95.8**	<u>94.3</u>	50.4	49.8	50.8	26.32%
		ProjectedGAN	**93.4**	<u>91.0</u>	51.2	49.8	50.7	26.52%
		Diff-ProjectedGAN	**92.7**	<u>90.2</u>	51.9	49.8	51.1	25.51%
		Avg.	**93.95**	<u>91.82</u>	51.12	49.75	50.85	26.07%
LAION & SDXL	LSUN-B	StyleGAN	**96.8**	<u>96.3</u>	50.7	49.8	50.9	12.16%
		ProjectedGAN	**94.8**	<u>93.8</u>	52.8	49.8	50.9	16.13%
		Diff-ProjectedGAN	**93.1**	<u>92.1</u>	53.6	49.8	50.9	12.66%
		Avg.	**94.88**	<u>94.07</u>	52.37	49.75	50.88	13.76%
Total average			**94.42**	<u>92.94</u>	51.74	49.75	50.87	20.90%

Table 4. Accuracy (%) on various evaluation datasets when trained on LAION & Kan3.

Eval set	Gen. model	Accuracy (%)		
		DID	Δ^2	DIRE
LSUN-B	ADM	86.5	69.0	84.0
	PNDM	83.0	81.5	67.5
	DDPM	82.0	78.0	71.5
ImageNet	ADM	98.9	98.4	91.0
	SDv1	98.9	98.7	93.5
LAION	Kan.3	99.9	100	99.9
	SDXL	99.9	99.9	99.2
	Vega	99.9	99.9	98.8

images. Second, in settings where LSUN is used as the evaluation dataset and synthetic images are generated by ADM, DIRE outperforms Δ^2. A closer inspection reveals that this corresponds to a simple scenario in which the synthetic images are produced by the same model used for reconstruction. This is consistent with our analysis in Section 3, which shows that first-order differences are sufficient in simpler settings. Finally, DID outperforms or performs comparably to both Δ^2 and DIRE across all settings, illustrating the advantages of leveraging both first- and second-order differences for classification. Additional ablation studies are reported in Section C.4.

5. Discussion

This paper studies the detection of AI-generated images. A popular approach is reconstruction-based, which relies on the first-order difference between an input image and its reconstruction as the basis for classification. Although these algorithms perform well when real and AI-generated images differ substantially, they often fail when the two appear highly similar. To address this challenge, we propose a difference-in-difference approach that utilizes a second-order difference. Our analysis in Section 3.3 shows that the second-order difference effectively removes perturbation noise and uncovers subtle discrepancies between real and AI-generated images. We then integrate both first- and second-order differences into our classifier to simultaneously handle both settings. Extensive experiments demonstrate that our detector outperforms state-of-the-art algorithms (Section 4.2); a sensitivity analysis shows that its performance is robust across different generative models (Section 4.3); and an ablation study confirms the critical role of the second-order difference (Section 4.4).

We focus on using second-order differences for detection, but our idea naturally extends to even higher-order differences. For example, one may define a third-order difference as

$$\Delta^3(x) = |\Delta^2(x) - \Delta^2(x')|, \tag{10}$$

for an imput image x and its reconstruction x'. In principle, one could incorporate all higher-order differences into the classifier to capture weak signals. However, computing a Kth-order difference requires reconstructing the image K times, which significantly increases the computation. Thus, there is an inherent trade-off: higher-order differences may uncover more subtle discrepancies between real and AI-generated images, but at the cost of increased computation. Determining the optimal order that balances detection performance and computational cost remains an open question for our future work.

Finally, although DID is proposed for image detection, the same principle can be readily generalized to the detection of LLM-generated text [see e.g., 8, 42, 53, 96, 98].

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